pfi

Python for Fixed Income

COURSE 3 · MARKET DATA AND TIME SERIES

Case Study: The Market Dataset

WEEK 1 · DAY 5

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The last day of week 1

Hands-on: a cache

write_cache, read_cache, and cached_fred_series: a series downloaded at most once a day, and a licensed series never stored.

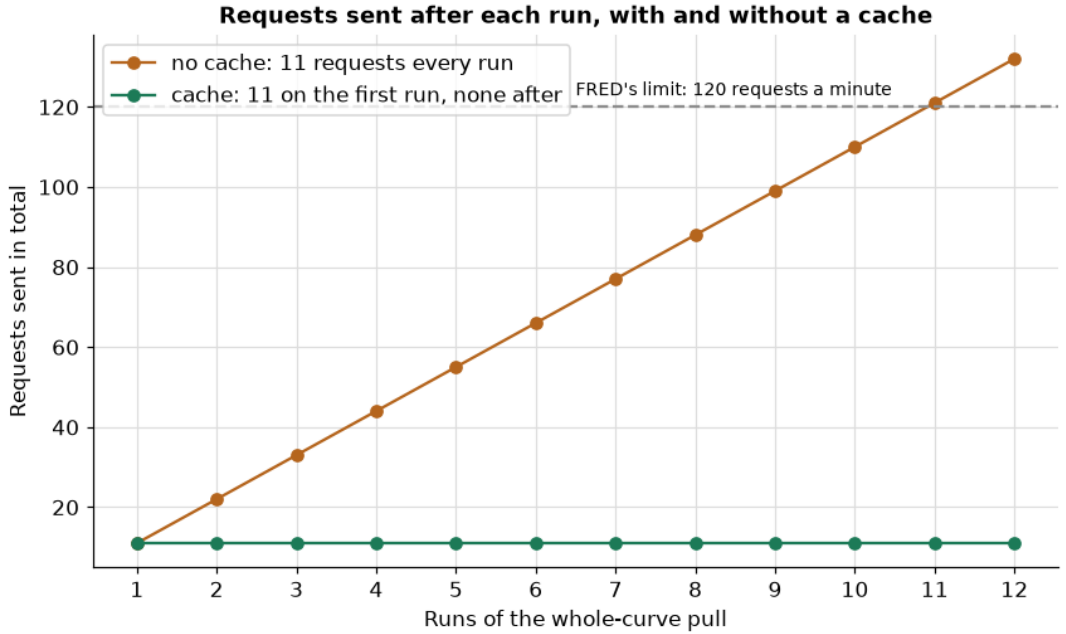
Case study: the market dataset

Five Treasury tenors in percent and two spreads in bp, one row per market day, checked with every test of week 1.

The spreads are synthetic: invented for this course, not ICE, Moody's, or any index.

No key? Every cell still runs. Live cells print the skip message, and that is the saved notebook.

Why cache



11 requests a run, 132 for twelve quick reruns. FRED's limit: 120 a minute, then error 429.

FRED API documentation, errors page: "up to 120 requests per minute" (read 2026-10-04).

Speed

A file on disk opens faster than a request crosses the internet and back.

Offline

On a train, or when the server is down, the copy is still there.

Fresh = today

A cached file written today is read; otherwise one request, then saved.

write_cache and read_cache

```
path = md.write_cache(ten, work / "cache")
# cache/treasury_10yr.csv, every digit kept
back = md.read_cache("treasury_10yr", cache)
back.equals(ten)    # True
```

A cache must give back exactly what it was given: a number one digit different in the 16th place is a different number.

The 10 Yr divided by 3, cached

pandas' default reader: 599 of 2,940 wrong

read_cache (round_trip): 0 of 2,940 wrong

Same file both times: the reader made the difference.

```
Missing file: no cache file DGS2.csv in the folder cache
```

Two tests: the round trip, exact for the FRED-format sample and for thirds; the missing file. 19 passed.

cached_fred_series: refuse first

1. Licensed?

ID starts with LICENSED_PREFIXES: ValueError, nothing sent

→

2. Fresh?

Today's file in cache/: read it, no request

→

3. Otherwise

fred_series, then write_cache, once a day

Series ID	In the cache?
DGS10, DGS2	may be cached (public domain)
BAMLC0A0CM, BAMLH0A0HYM2	refused: ICE BofA
BAA10Y, DBAA, DAAA	refused: Moody's

Freshness: written today True; moved back a day False. Third test: no request, no folder. 20 passed.

ICE and Moody's on FRED: their terms

ICE BofA (BAMLC0A0CM notes)

"...provided for your internal use only and you are not authorized or permitted to publish, distribute or otherwise furnish Top Level Data to any third-party without prior written approval of ICE Data."

Moody's (BAA10Y notes)

"MOODY'S INFORMATION MAY NOT BE COPIED OR OTHERWISE REPRODUCED, ... OR STORED FOR SUBSEQUENT USE FOR ANY SUCH PURPOSE ... WITHOUT MOODY'S PRIOR WRITTEN CONSENT."

"Starting in April 2026, this series will only include 3 years of observations." ICE BofA notes on FRED

Never cached, never written, never on screen: live cells print metadata only, values only under SHOW_LICENSED_VALUES, which stays False.

fred.stlouisfed.org/series/BAMLC0A0CM and /series/BAA10Y, series notes (read 2026-10-04).

Case study: the market dataset

Every morning a desk reads the curve and spreads from one daily dataset. A weekend row, a lost holiday, a blank tenor, or a spread on the wrong day spoils every number built on it.

Objective: build it, and prove it clean

Dates

unique, ascending, no weekends

Calendar

missing weekdays = 126

Late tenors

found and described

Spreads

on every market date

Stale

no run of 5 in a spread

File	Holds	Units	Licence
treasury_par_yields.csv	2,940 market days, 14 tenors	percent	Public domain
course3_synthetic_spreads.csv	ig_spread_bp, hy_spread_bp	bp	Synthetic, MIT

The curve, checked

2,940

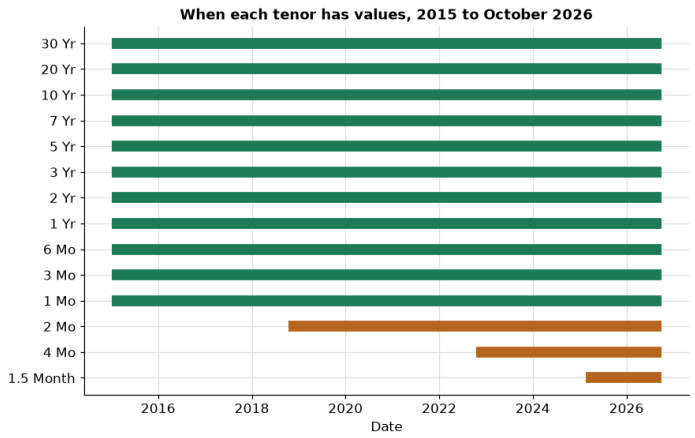
dates, unique and ascending, none on a weekend

126

weekdays with no row: the same as day 3

3

tenors that start late, with no gap after



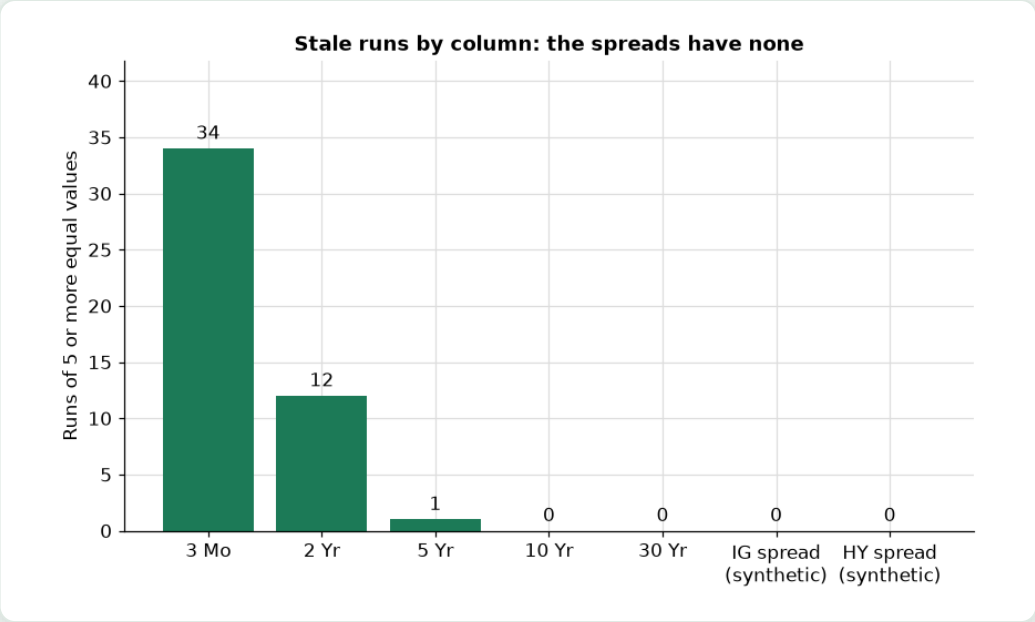
Tenor	First value	Blank before	Gaps
1.5 Month	2025-02-18	2,532	0
2 Mo	2018-10-16	949	0
4 Mo	2022-10-19	1,951	0

A blank before a tenor starts is not a missing value to fill: there was nothing to publish.

The spreads, joined

```
inner = md.align({...}, how="inner")
outer = md.align({...}, how="outer")
# 2,940 rows both ways, nothing missing
```

Same rows both ways: a spread on every market date, and none on a date the curve lacks.



The stop is on the spreads. Yields repeat at two decimals: reported, not removed, since they are what Treasury published.

The dataset, checked

date	3 Mo	2 Yr	5 Yr	10 Yr	30 Yr	ig_spread_bp	hy_spread_bp
2026-09-30	4.20	4.88	5.09	5.29	5.64	98.8	334.7

2,940 rows, 2015-01-02 to 2026-10-02. Yields in percent, spreads in bp (synthetic). Every week 1 check silent.

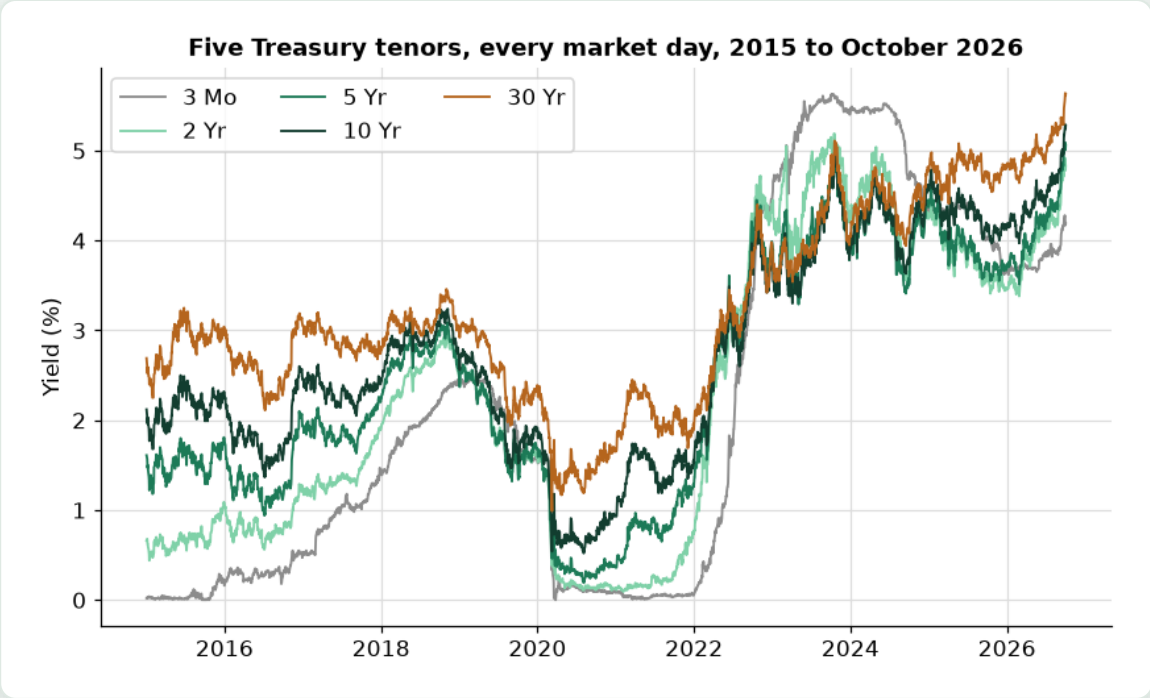
The 10 Yr is the month-end value day 4 pinned; the spreads are the synthetic book's weighted spreads that day.

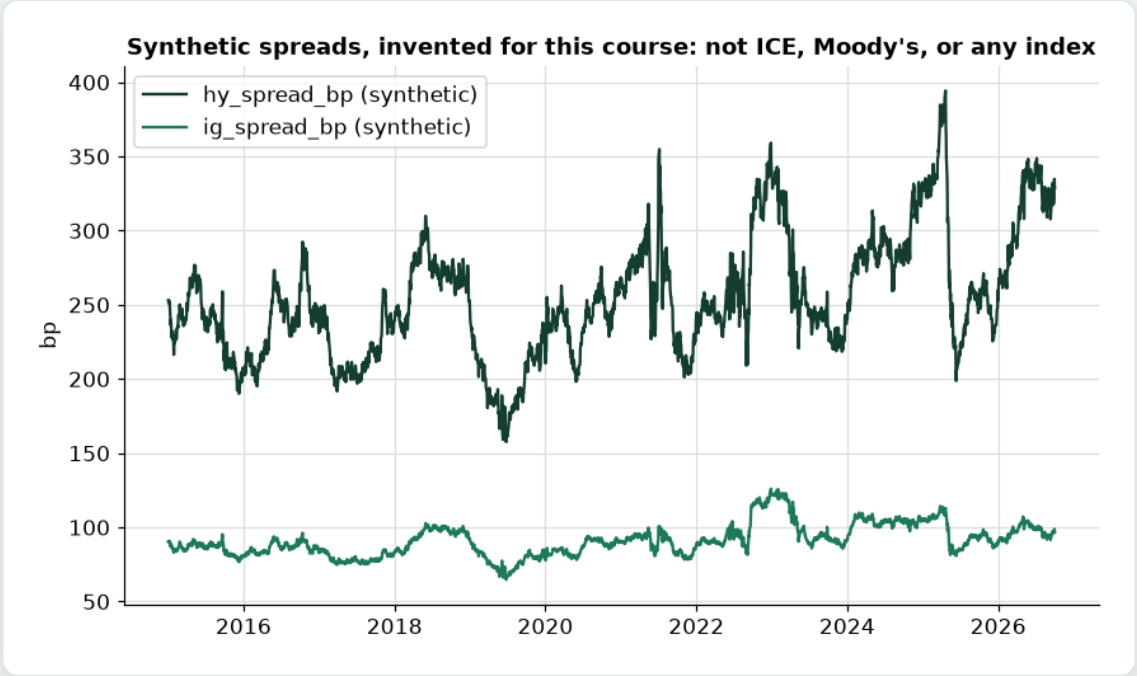
Observations: they describe

High yield above investment grade on every row; no stale run of 5 in either spread; yields repeat and are kept. The spreads are invented, so they say nothing about any real market.

Nothing here says where any yield or spread is going.

The dataset, drawn





Seven columns, every market day, no blank. The spreads are synthetic: invented, not ICE, Moody's, or any index.

Two live cells

11 DGS series, cached

cached_fred_series into cache/, each checked against the snapshot to a billionth of a percent. A second pass reads 11 files written today: no request.

ICE and Moody's: metadata only

fred_series into memory. Prints ID, units, rows, first and last date, dot rows, weekend rows. Checks percent, about three years for ICE, nothing cached.

A live cell, two ways

No key (the saved notebook):

```
Live cell skipped: no FRED key was found.
```

With your key:

eleven checked series; for ICE and Moody's, counts and dates only

Excel side by side: Append and Merge

Append (Power Query)

`Table.Combine({First, Second})`

The Treasury file split at 2026: 2,750 + 190 rows. Appended: 2,940 rows, 15 columns, equal to the file.

`Python: pd.concat([first, second])`

Merge (Power Query, left outer on date)

`Table.NestedJoin, then Table.ExpandTableColumn`

2,940 rows, 17 columns. Spreads arrive as text ("98.8", "334.7") until a step sets type number.

`Python: md.align({...}, how="inner")`

Data, Get Data, Combine Queries, Append or Merge.

Tested in Excel 16.0 build 20527, 2026-11-02.

Excel side by side: XLOOKUP and the checks

Excel side by side (tested in Excel)

`=XLOOKUP($A2,Spreads!$A$2:$A$2941,Spreads!$B$2:$B$2941)`	0 #N/A in 2,940 rows
`=SUMPRODUCT(--(WEEKDAY(A2:A2941,2)>5))`	0
`=ROWS(A2:A2941)-ROWS(UNIQUE(A2:A2941))`	0
`=SUMPRODUCT(--(A3:A2941<=A2:A2940))`	0
`=COUNTBLANK(F2:F2941)`	1,951
`=XLOOKUP(TRUE,F2:F2941<>"",A2:A2941)`	2022-10-19

The same answers as Python: weekends, repeats, order, the 4 Mo blanks, and its first date.

Save the day: tag week 1

1. Rebuild

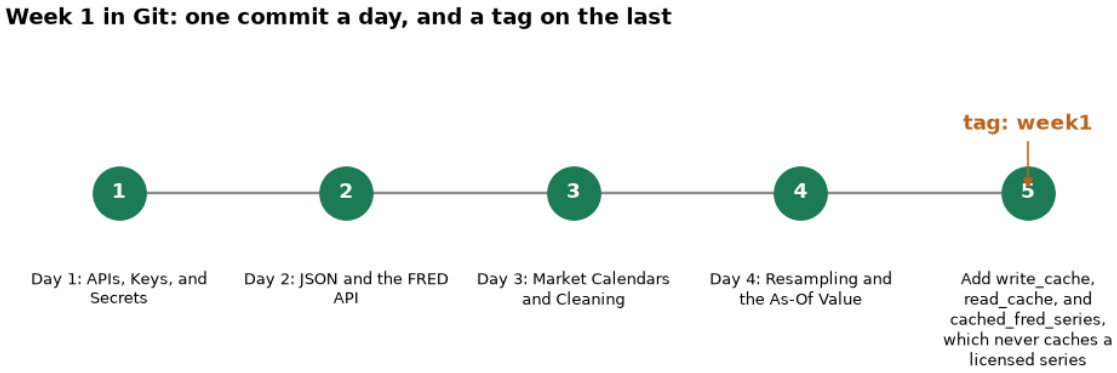
Days 1 to 4 from today's files: each day is the front part of the next

2. Commit

Today's three functions and three tests

3. Tag

git tag week1: a name fixed to this commit



git log --decorate shows (HEAD -> main, tag: week1) beside today's commit. cache/ stays ignored: git check-ignore -v names .gitignore:4:cache/

The AI check

The docstring is the prompt. The code that comes back runs, and has one data bug.

1

Docstring

Units, conventions, and what it must never do: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: dates put on New York time "to be safe" move back a day: the first is 2015-01-01, and 2026-09-30 holds 1 October.

Code that runs is not the same as code that is right.

What you can do now

- Cache a series once a day, and read it back exactly
- cached_fred_series refuses a licensed series before sending anything
- ICE and Moody's terms: their values never appear in this course
- The market dataset, proved clean with every check of week 1
- Tag the end of a week: git tag week1

Week 2, Day 1 Changes in bp and percent: yields and spreads in bp, prices in percent, and the as-of value